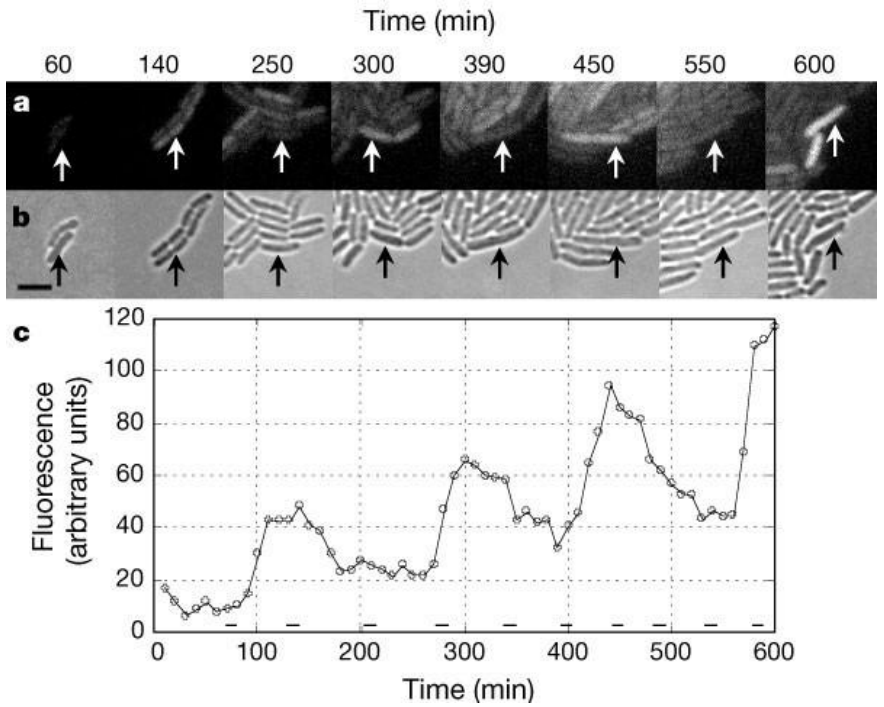
Can we predict all phenotypes if we know or control all genes and all environments?

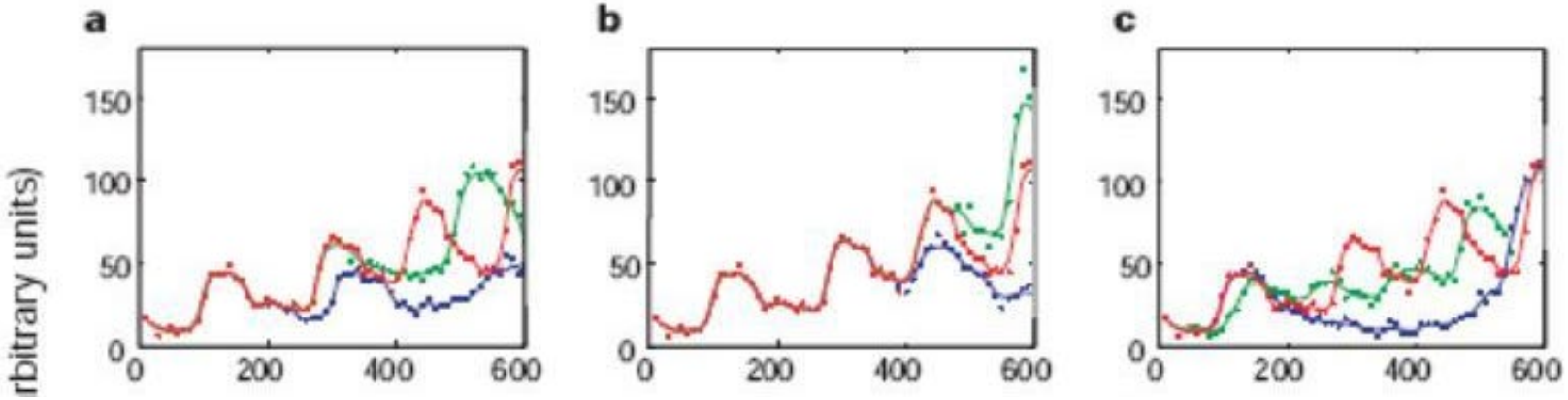
Can we predict the fate of individual (cells or people) if we know everything about the?

The repressilator worked



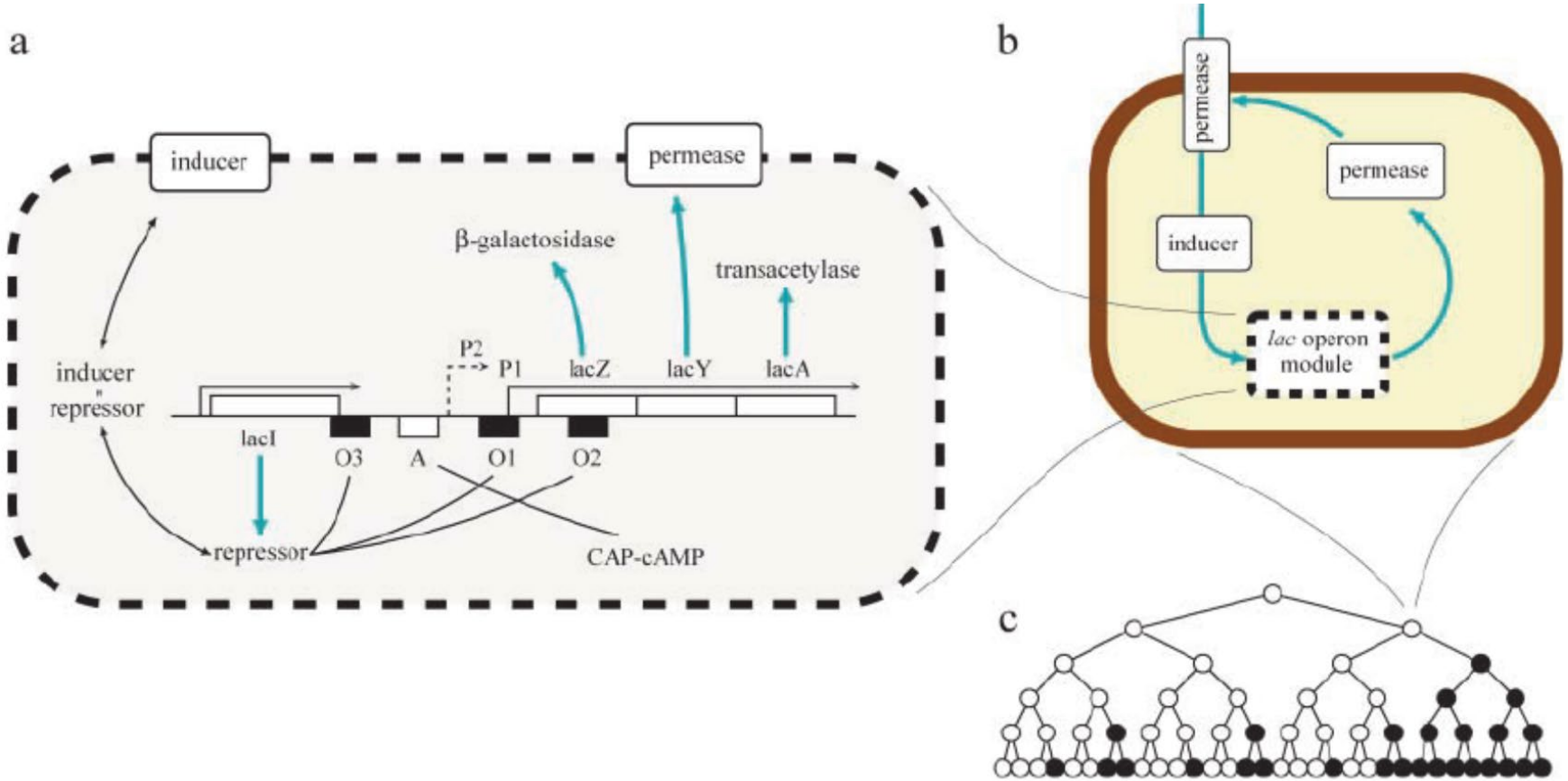
Elowitz MB & Leibler S, Nature (2000).

But it was noisy



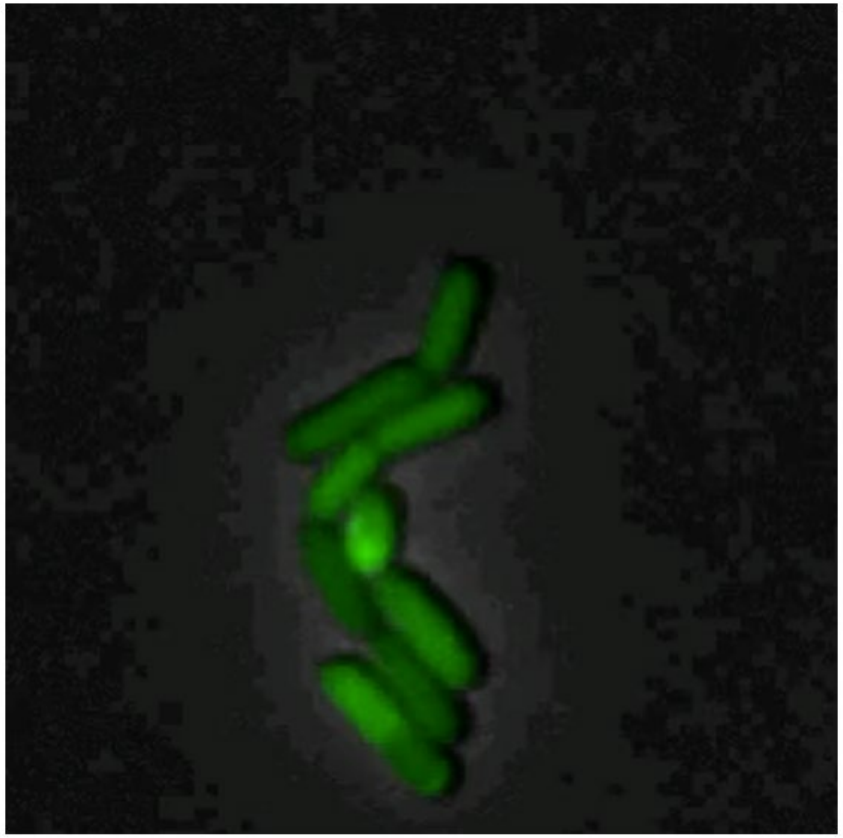
Elowitz MB & Leibler S, Nature (2000).

What accounts for the switching between bistable states created by positive feedback?



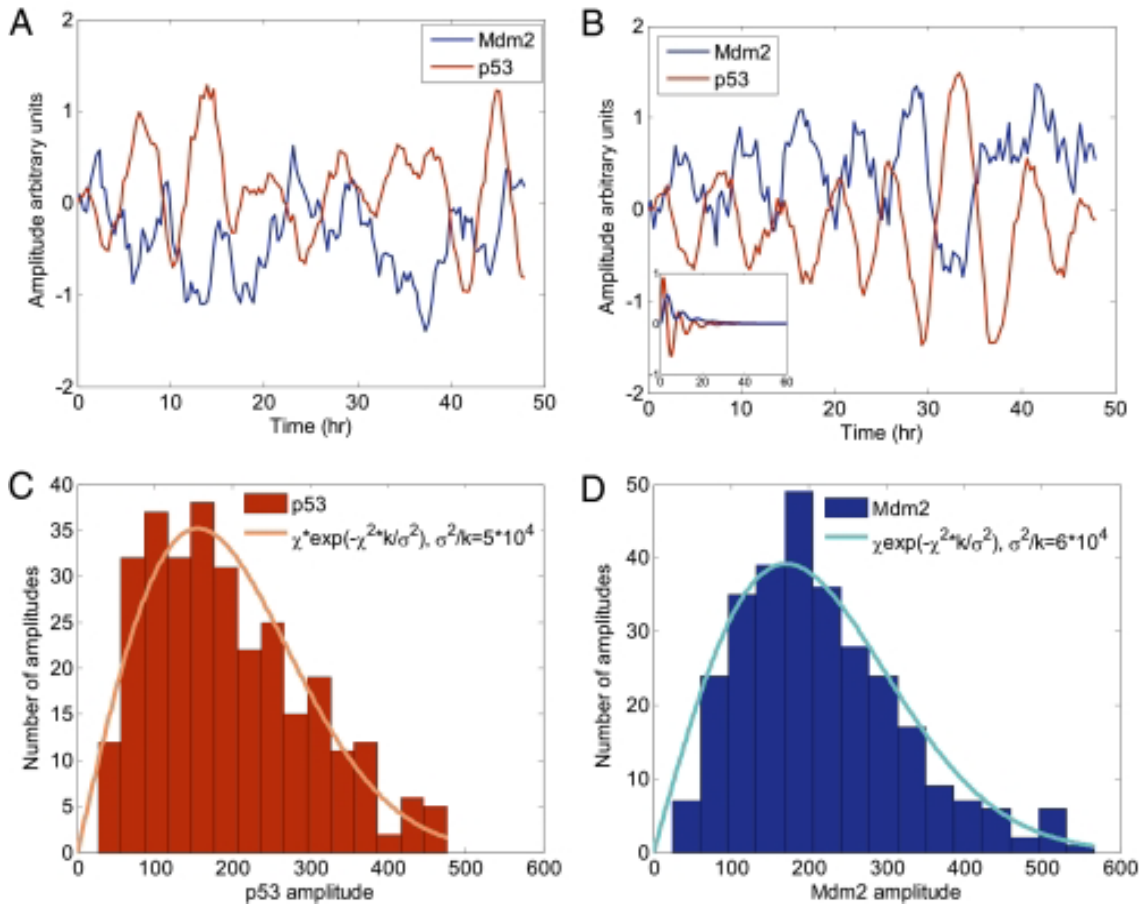
Vilar, Guet & Leibler, JCB (2003)

Clonal cells in a homogeneous environment can adopt different developmental fates



Locke JCW, Elowitz MB Nature Rev. Microbiol. (2009)

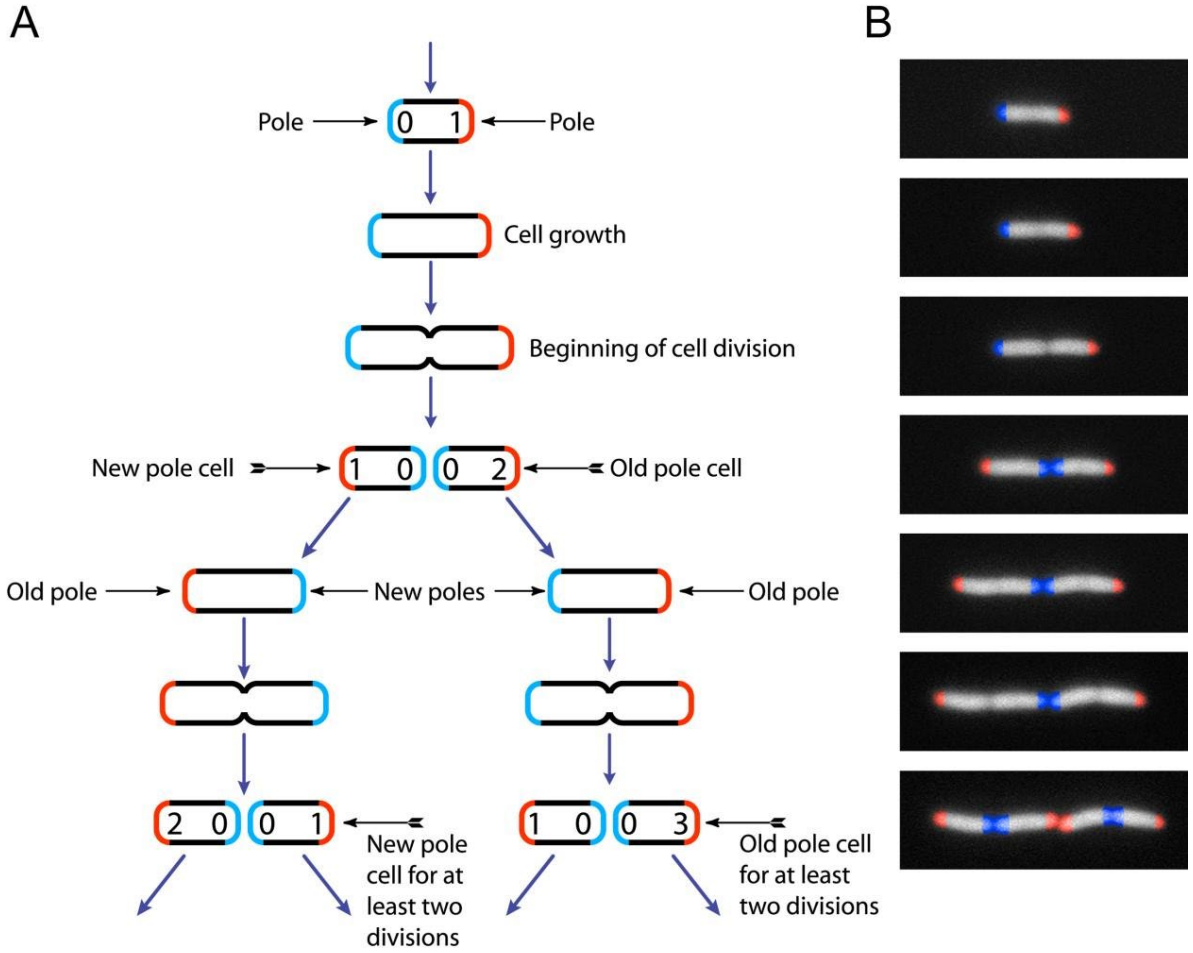
We even think noise can sustain oscillations around a stable spiral



Geva-Zatorsky et al., PNAS (2010)

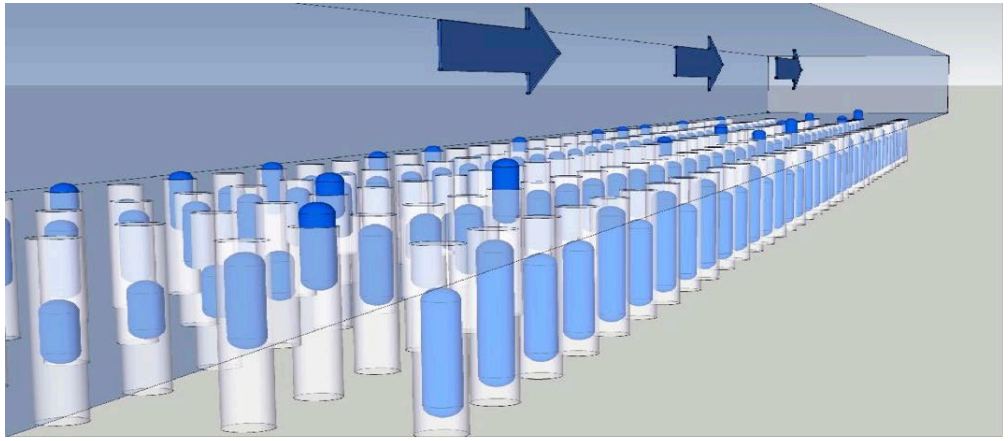
What is noise mechanistically?
Is noise just something we haven’t measured yet?

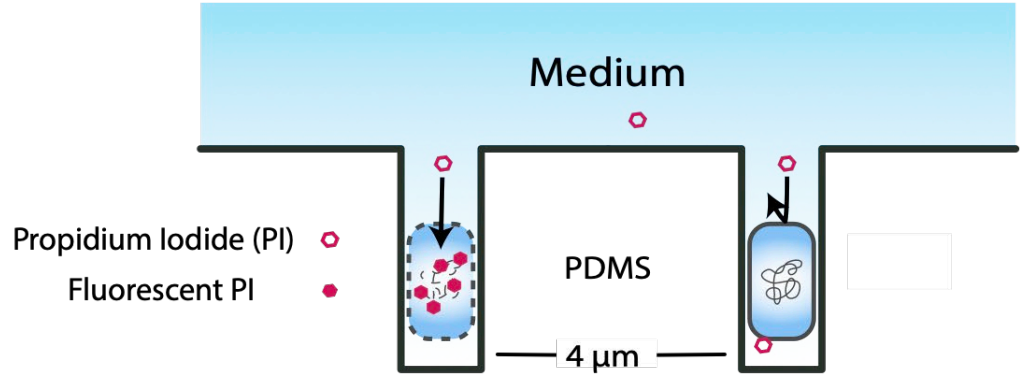
Cells are never identical: there are hidden differences even among sister cells



Stewart E et al PLoS Biology (2005)

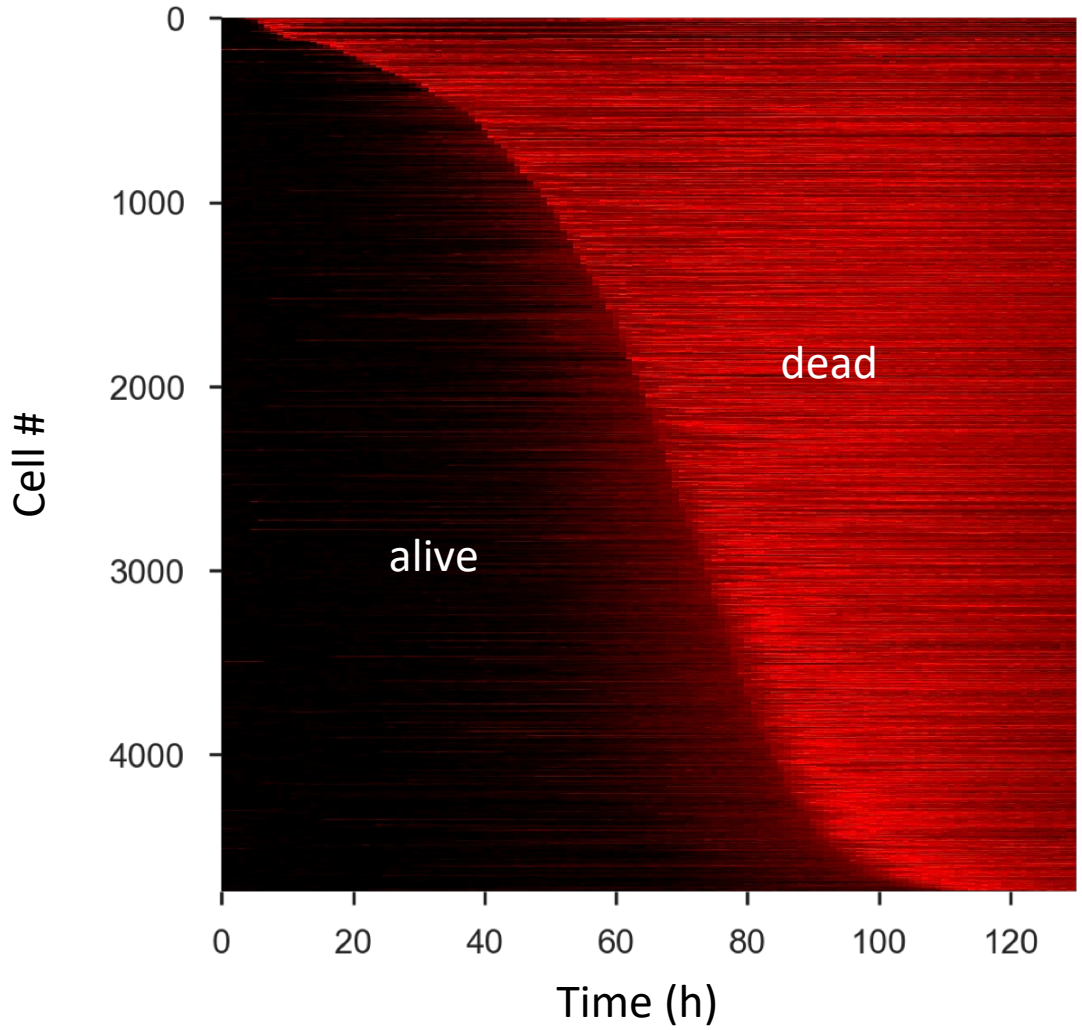
Genetical, environmental and historical variability can be controlled in quantitative microbial experiments





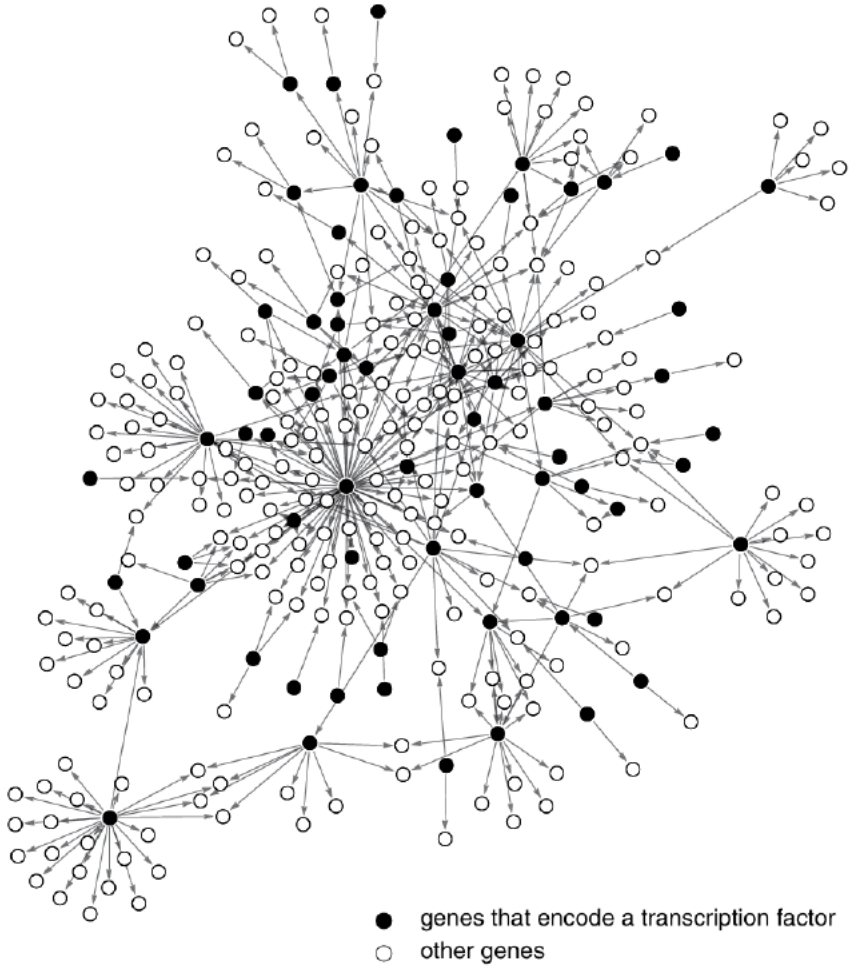
Yang Y et al (2019) Science Advances

Genetically identical cells in homogenous environment die at different times



Yang Y et al (2019) Science Advances

Noise can propagate across chemical reaction and gene regulatory networks

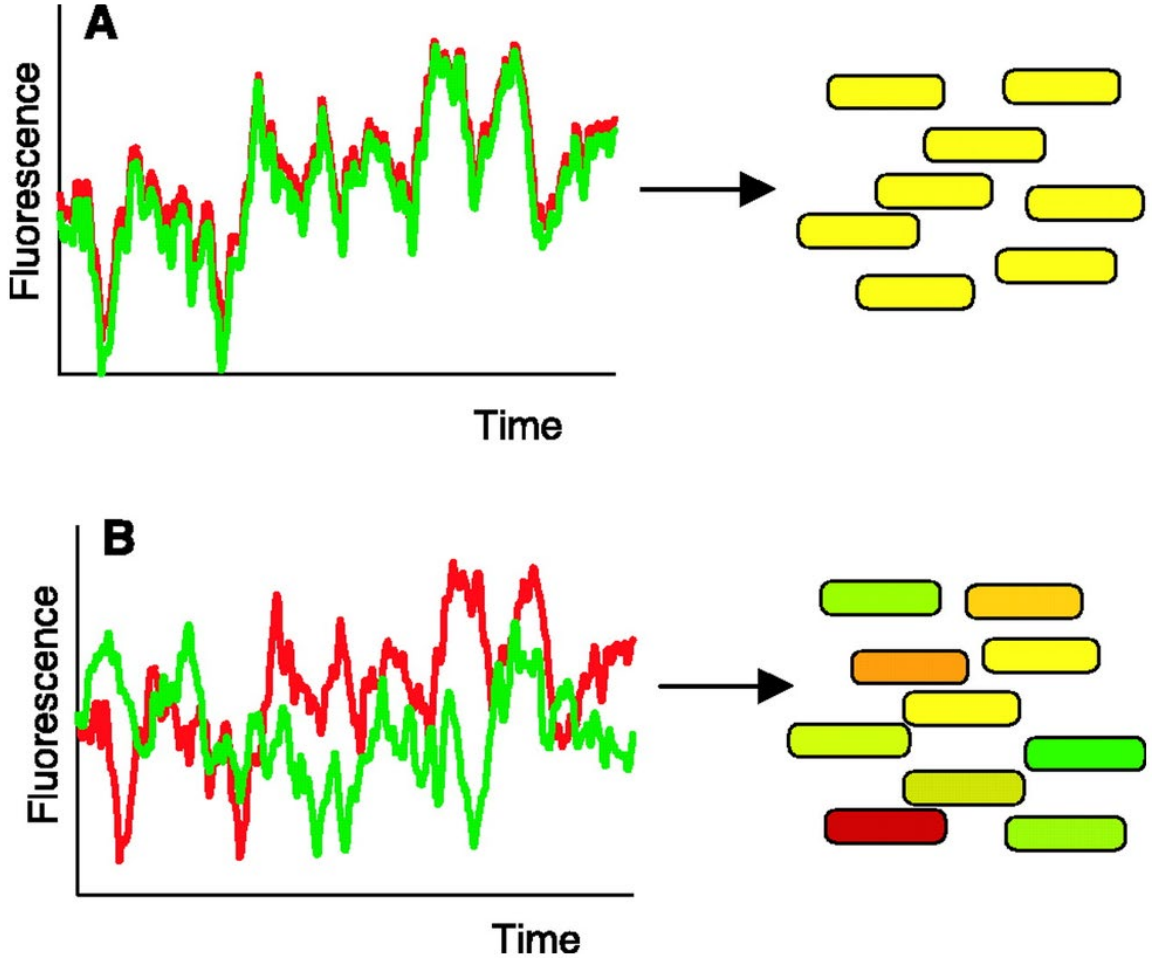


- N=4,500 nodes-genes
- E = 10,000 arrows

Uri Alon (2019) An Introduction to Systems Biology 2nd edition

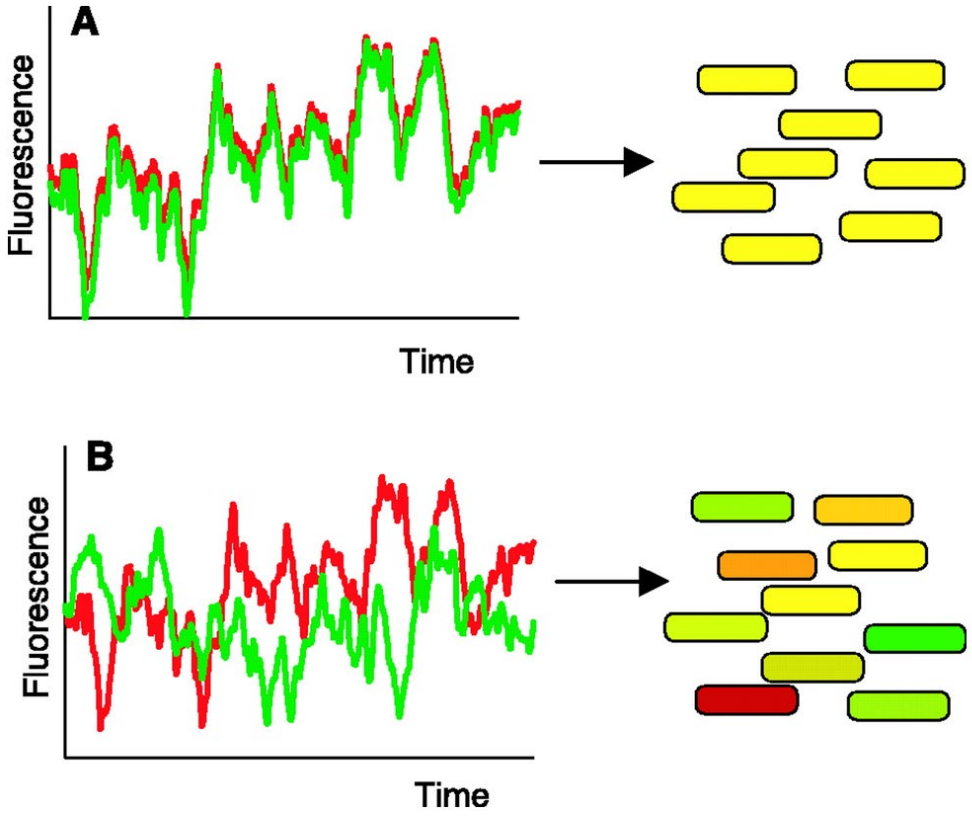
How can we prove stochasticity in biology?

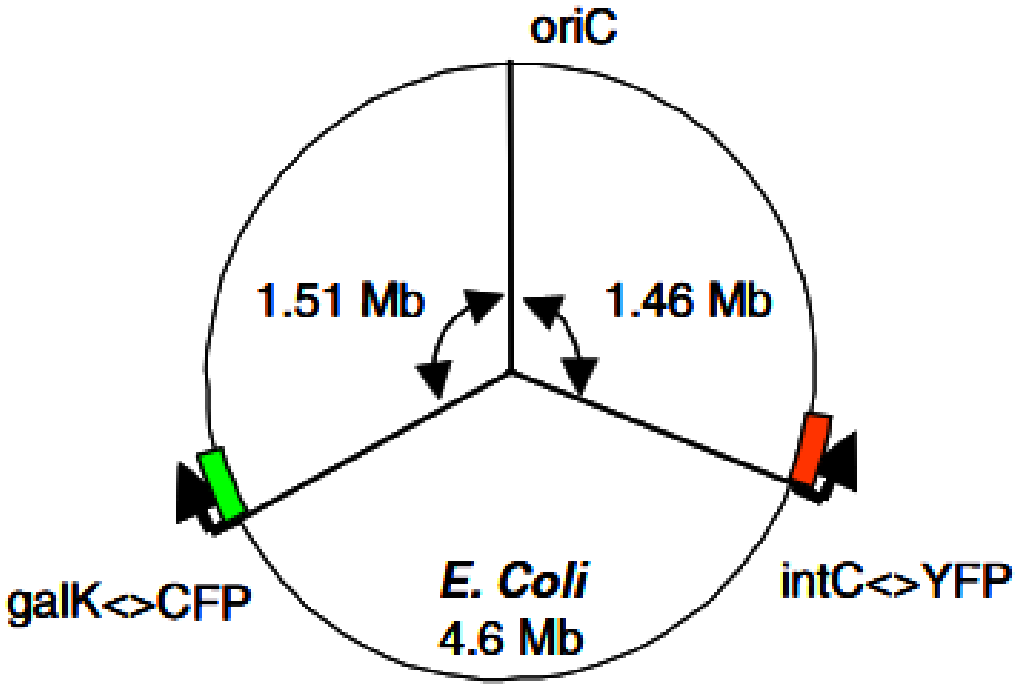
If we identical genes are place in the same fluctuating cell, do they response similarly?



Elowitz MB, Levine AJ, Siggia ED, Swain PS. Science (2002)

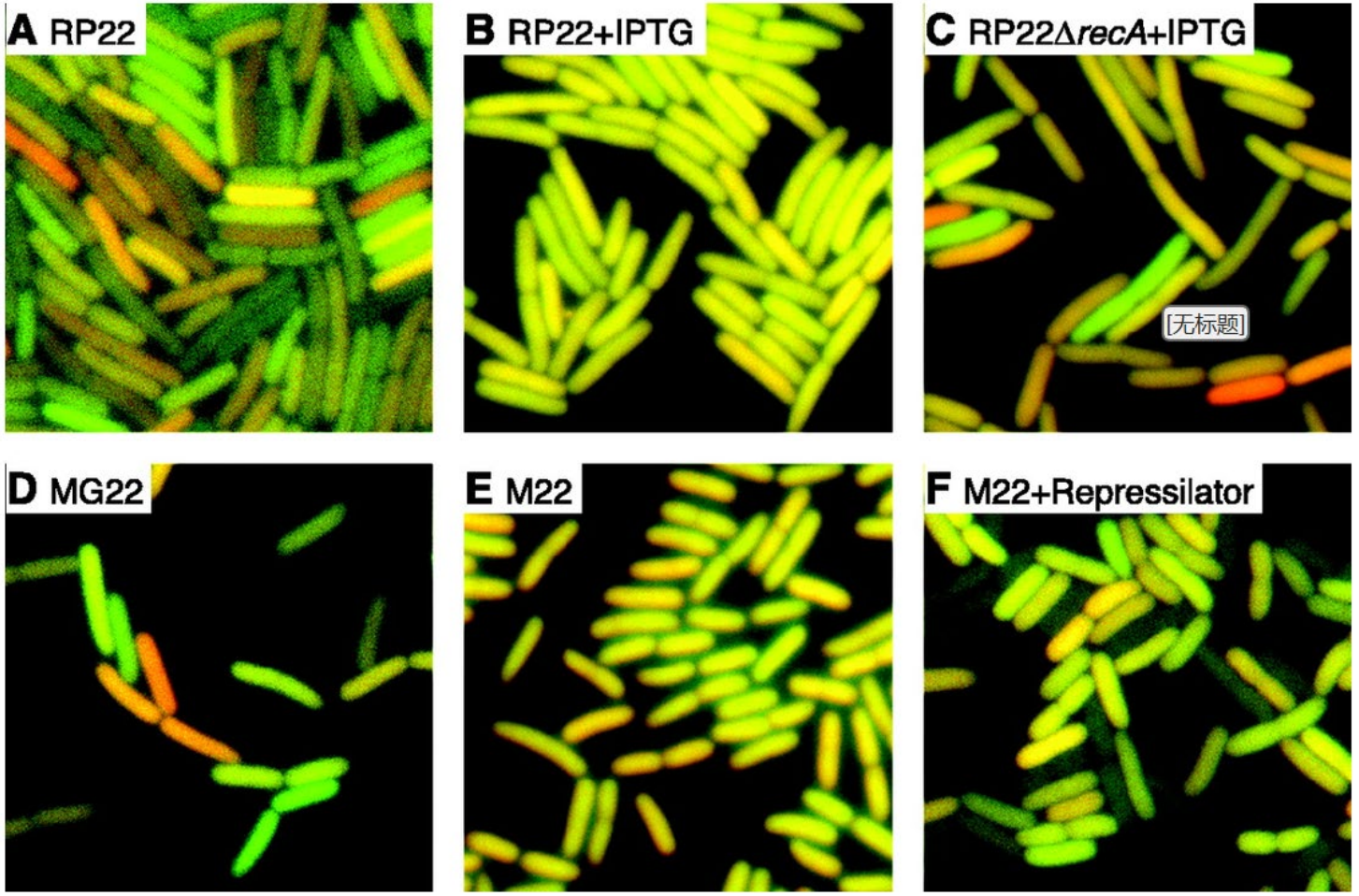
Elowitz et al placed 2 genes differing by 2 point-mutations in similar locus on the E coli genome





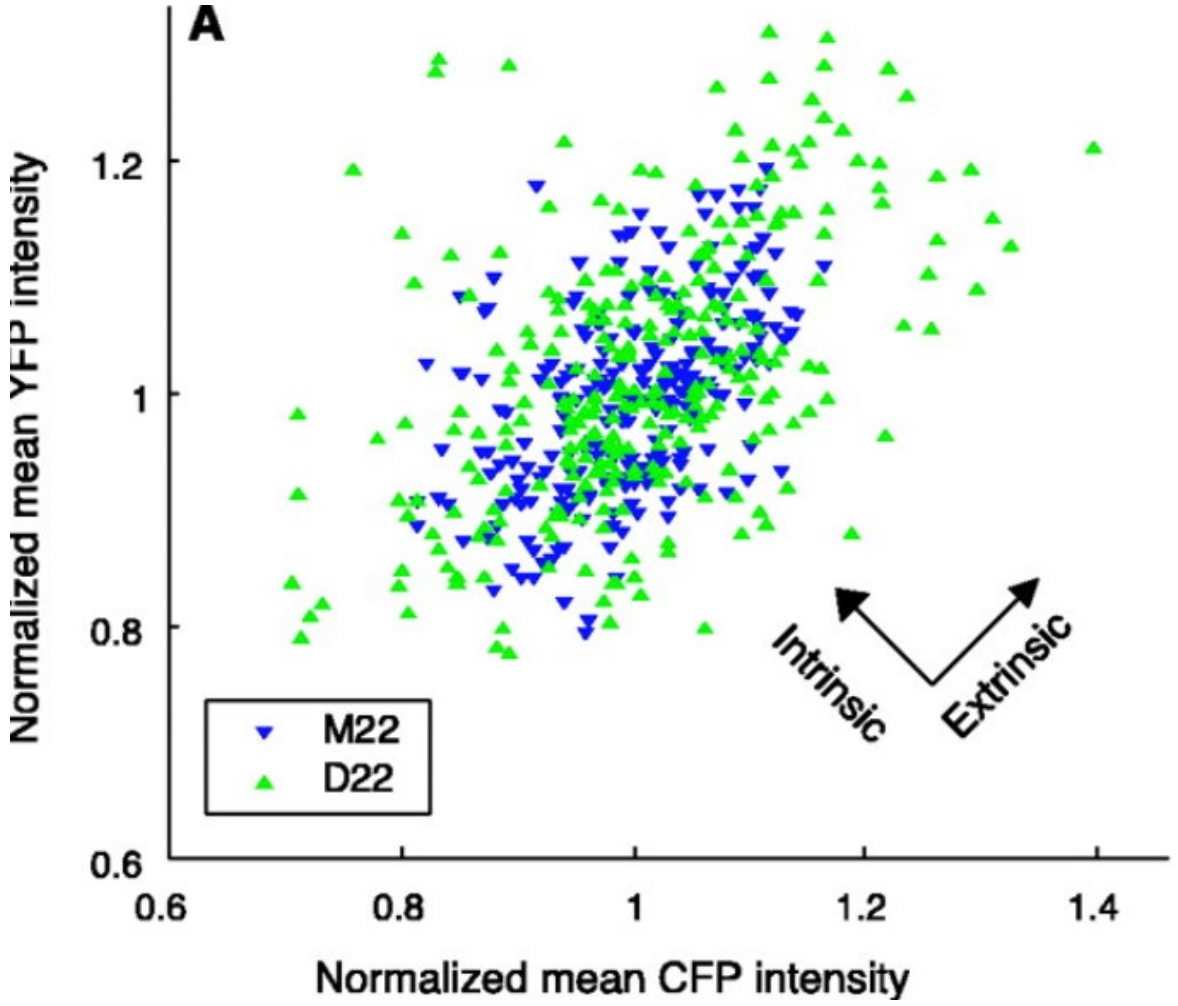
Elowitz MB, Levine AJ, Siggia ED, Swain PS. Science (2002)

Even identically regulated genes can exhibit different expression levels within the same cell



Elowitz MB, Levine AJ, Siggia ED, Swain PS. Science (2002)

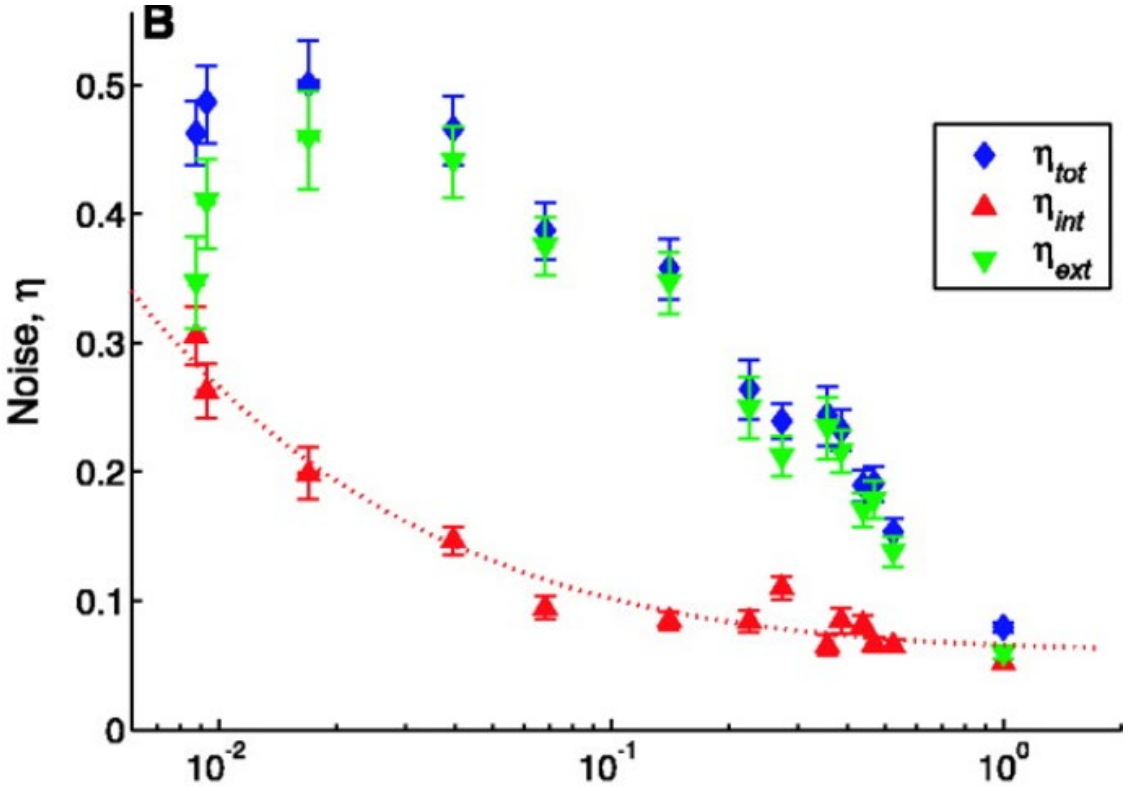
Correlations and differences between 2 nearly identical genes can separate intrinsic noise from propagation of noise elsewhere in the GRN



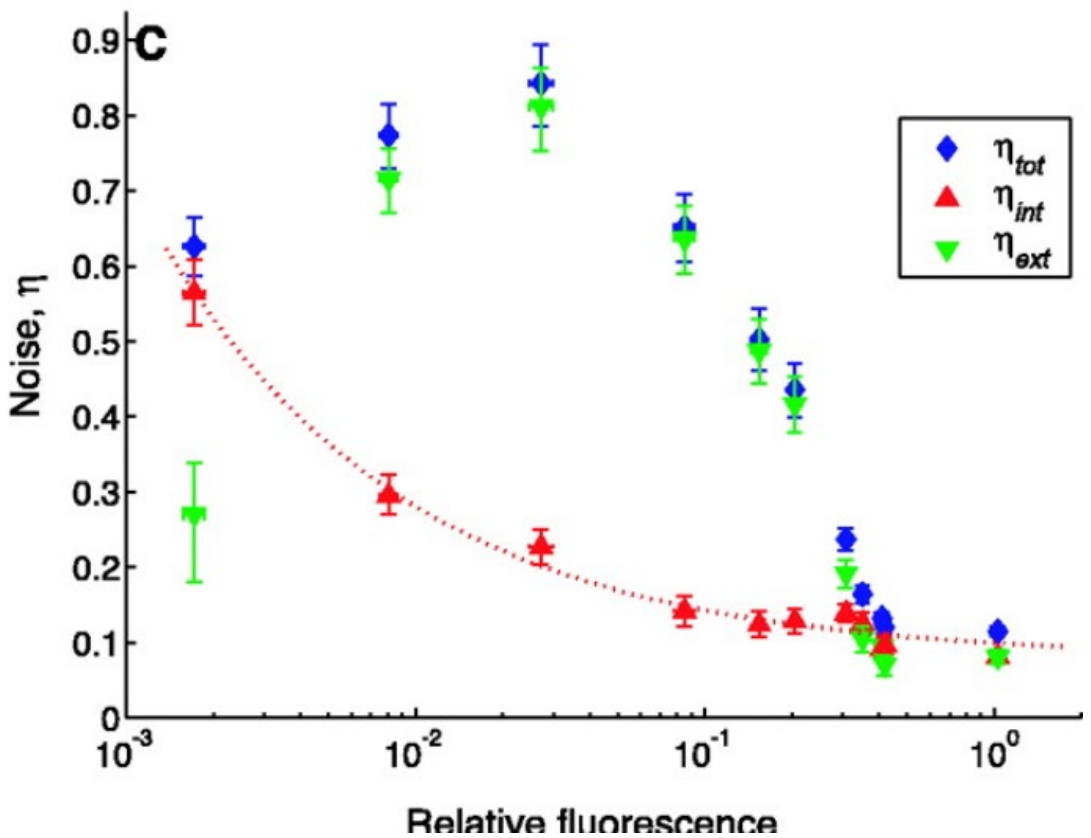
Elowitz MB, Levine AJ, Siggia ED, Swain PS. Science (2002)

Intrinsic noise can be shown to scale with the inverse square root of the expression level

recA+



ΔrecA



Elowitz MB, Levine AJ, Siggia ED, Swain PS. Science (2002)

Chemical master equations provide a mathematical framework to derive noise properties analytically

$$\frac{\sigma_2^2}{\langle n_2 \rangle^2} \approx \overbrace{\frac{1}{\langle n_2 \rangle H_{22}}}^{\text{Intrinsic } n_2 \text{ noise}} + \overbrace{\underbrace{\frac{\sigma_1^2}{\langle n_1 \rangle^2}}_{n_1 \text{ noise}} \underbrace{\frac{H_{21}^2}{H_{22}^2}}_{\text{Susceptibility}} \underbrace{\frac{H_{22} \,/\, \tau_2}{H_{11} \,/\, \tau_1 + H_{22} \,/\, \tau_2}}_{\text{Time-averaging}}}^{\text{Extrinsic } n_2 \text{ noise}} \quad (1)$$

Paulsson, Nature (2004)